SUPPLY CHAIN ANALYTICS

END-TO-END DATA ANALYST PROJECT

Excel • SQL Server • Power BI • Python

PROJECT REPORT

An end-to-end analysis of supply-chain operations, sales, profitability, shipping performance, and late-delivery risk. The workflow combines spreadsheet analysis, SQL-based analysis, interactive Power BI dashboards, and Python exploratory data analysis.

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Portfolio-ready summary for GitHub. Findings reflect the results reported during the project workflow; definitions and limitations are stated below.

65,752	180,519	$36.78M	$3.97M
Distinct orders	Dataset rows	Total sales	Total profit

Project at a glance

- **Dataset:** DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS.
- **Analysis grain:** 180,519 rows and 65,752 distinct Order Id values. Rows and distinct orders are not interchangeable.
- **Business focus:** sales and profit contribution, category performance, shipping duration, delivery status, and late-delivery risk.
- **Tools:** Excel, SQL Server, Power BI, Python (pandas, NumPy, Matplotlib, Seaborn).

1. Business Objective

Build a multi-tool analytics project that turns a supply-chain dataset into useful commercial and operational insights. The analysis examines sales and profit across markets and categories, shipping performance, and late-delivery risk.

Project workflow

01 · Excel	02 · SQL Server	03 · Power BI	04 · Python
Prepare data, calculate KPIs, create pivot-based views and an overview dashboard.	Query data, calculate KPIs, create reusable views for market, category, and shipping analysis.	Build three dashboards: executive overview, category profitability, and market/operations.	Clean fields; perform EDA, aggregations, visualizations, risk analysis, and trend analysis.

2. Tools and Technical Work

Tool	Work completed
Excel	Data preparation, KPI formulas, pivot tables, slicers, and an overview dashboard, including orders, sales, profit, margin, and shipping status.
SQL Server	Imported the CSV dataset, checked and transformed relevant fields, calculated metrics, and created reusable views for detailed, KPI, category, and market analysis.
Power BI	Three dashboards: Executive Supply Chain Overview; Category Profitability & Operations; Market & Operational Performance.
Python	pandas and NumPy for data handling and calculations; Matplotlib and Seaborn for charts; market/category analysis, shipping-risk analysis, monthly trends, correlations, and validation.

3. Key Results and Findings

Overall metrics

Metric	Result	Definition / interpretation
Distinct orders	65,752	Count of unique Order Id values.
Dataset rows	180,519	Rows are not equivalent to unique orders.
Total sales	$36,784,734.31	Python result; compare using the same column, filters, and aggregation in each tool.
Total profit	$3,966,902.97	Sum of Order Profit Per Order at row level.
Profit margin	10.78%	Total profit divided by total sales.
Late-delivery percentage	57.29%	98,977 late-delivery rows divided by 172,765 non-cancelled rows; rounds to 57%.

Market and category findings

- **Highest-sales market:** Europe, with reported sales of $10.87 million.
- **Most profitable category:** Fishing, with total profit of $756,220.76 and calculated profit margin of 10.91%.
- **Highest date-calculated delayed-shipment share by market:** Pacific Asia at 35.71%. This uses a date-derived classification and is different from the original Delivery Status late percentage.
- **Monthly sales extremes:** October 2017 had the highest sales ($895,572.52); January 2018 had the lowest ($121,139.25). Review seasonality and data coverage before interpreting the difference.
- **Correlation:** Operational delay days and the original late-delivery-risk flag showed a positive correlation of approximately 0.78. Correlation does not prove causation.

Shipping metric definitions

The notebook uses two approaches: (1) original Delivery Status, where late-delivery rows divided by non-cancelled rows gives 57.29%; and (2) a date-derived classification comparing calculated shipping duration with scheduled duration. The date-derived result included 71,103 Unknown rows. These metrics are not interchangeable; investigate missing or unparseable dates before drawing conclusions from the date-derived classification.

4. Dashboards and Deliverables

- **Dashboard 1 — Executive Supply Chain Overview:** headline KPIs and an overall view of sales, profit, and delivery performance.
- **Dashboard 2 — Category Profitability & Operations:** category sales and profit analysis, profitability comparisons, and operational indicators.
- **Dashboard 3 — Market & Operational Performance:** market-level commercial and shipping performance comparisons.
- **Excel overview dashboard:** KPI cards and pivot-based charts with year/month filtering.
- **Python notebook:** data cleaning, EDA, charts, shipping-risk analysis, monthly trends, correlation matrix, and validation checks.
- **SQL Server objects:** detailed analysis and KPI/category/market views supporting reporting.

5. Business Recommendations

- Investigate markets with high late-delivery shares; compare shipping mode, scheduled duration, and order mix before proposing interventions.
- Review category profitability alongside sales volume; investigate strong-selling categories with weak or negative profit.
- Use monthly trends for planning only after checking seasonality and data coverage.
- Keep KPI definitions consistent across tools, especially cancelled-order exclusions and original-status versus date-derived delivery measures.
- Add a data-quality check for missing or unparseable calculated shipping dates.

6. Suggested GitHub Repository Structure

supply-chain-analytics/
■■■ README.md
■■■ report/Supply_Chain_Analytics_Project_Report.pdf
■■■ python/SC_Analysis.ipynb
■■■ sql/analysis_queries_and_views.sql
■■■ dashboards/excel_dashboard.png
■■■ dashboards/powerbi_dashboard_1.png
■■■ dashboards/powerbi_dashboard_2.png
■■■ dashboards/powerbi_dashboard_3.png

Recommended repository assets: notebook, SQL scripts, dashboard screenshots, this PDF report, and a README explaining the dataset, methodology, KPI definitions, findings, and limitations. Avoid committing credentials or large raw files without checking dataset terms.

7. Limitations and Validation Notes

- Row count and distinct-order count represent different levels of aggregation.
- The sales and profit totals in this report are the reported Python values. Before publication, ensure each tool uses the same source column, row grain, filters, and aggregation; the user previously noted a different Excel sales total.
- The date-derived shipping classification contains 71,103 Unknown rows; investigate before treating it as a definitive delivery KPI.
- Dashboard screenshots were not embedded because they were not provided for this report-generation step.

Project conclusion

This project demonstrates a practical multi-tool analytics workflow, from preparing and querying data to building interactive dashboards and using Python for exploratory analysis. Its portfolio value comes from combining technical analysis with business-facing metrics and documenting KPI definitions and data-quality limitations.

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